Assignment 5: Restaurant Order System

Note: Assignment 5 is a double assignment. Each milestone (MS1 and MS2) is worth 1/6 of the assignments grade for the course, the same as (individually) Assignments 1–4.

Due:

- Milestone 1 due **Mon Apr 20th** by 11pm
- Milestone 2 due **Mon Apr 27th** by 11pm (no late hours allowed)

Note that you may **not** use late hours on Milestone 2. Please plan accordingly.

Updates:

4/14: updated the Updater Client and Display Client sections to clarify that

1. A `MessageType::ERROR` response to a login request should result in the client printing an error message to `std::cerr` and immediately exiting with a non-zero exit code
2. When the status of an order changes to `OrderStatus::DELIVERED`, it should be immediately removed from the display client's collection of orders, before the display is refreshed

4/15: linked to a screencast video with some suggestions for running and testing the client and server programs.

4/19: added content to the Server section, and added Synchronization of Shared Data, `MessageQueue`, Broadcasting to Display Clients, and Synchronization Report sections. These changes include all of the information you should need to complete Milestone 2 of the assignment.

4/20: fixed to clarify that the `IO::send` and `IO::receive` functions are declared in `include/io.h`.

Quick Guide

Here are the high-level steps we recommend for completing the assignment.

For Milestone 1:

1. Implement the `Wire::encode` and `Wire::decode` functions. Get all of the unit tests in the `message_tests` unit test program working. (See the Encoding section.)
2. Implement the `IO::send` and `IO::receive` functions. Get all of the unit tests in the `io_tests` test program working. (See the Framing section.)
3. Implement the `updater` client and test it

4. Implement the `display` client and test it

For Milestone 2:

1. Implement the `Server::server_loop` member function so that the server listens for TCP connections from clients. For each client that connects, create a `Client` object, and in a new detached thread, call its `chat` member function to communicate with the remote client.
2. Implement the `Client::chat` member function sufficiently that the client can log in. Test that you can log in successfully using the `nc` (netcat) program. You can also try using your client implementation or the reference client implementations, although they won't work fully until you implement the server functionality.
3. Add functionality to the `Client` and `Server` classes to implement the required server functionality. You'll need to add a central data structure to `Server` to keep track of orders, and use appropriate synchronization so that client threads can access the data concurrently.
4. Implement the protocol for communicating with an updater client. This will involve code in both `Server` and `Client`.
5. Implement the protocol for communicating with a display client. Each `Client` should have a `MessageQueue` object that the code in the `Server` object can use to post messages to be sent to the remote display client program which the state of any order or item changes. You'll need to implement the `MessageQueue::enqueue` and `MessageQueue::dequeue` member functions.

Milestone 2 tasks 3–5 will likely be the most challenging ones, although they should only involve a couple hundred lines of code, which should be fairly straightforward if you have thought about the problem and determined how to factor the problem into helper functions.

⚠️ This is Complicated, Help!

Specifying the intended behavior of a networked application entails some complexity. This assignment description aims to document everything you need to know to implement the server and clients for the restaurant order system.

The good news is that the assignment skeleton file (see Getting Started) includes reference executables for all three programs. You can use them as reference for how your programs should work. Also, you can use them to test your programs. For example, in Milestone 1, it will make sense to use the reference server implementation to test your client program implementations against.

Also, the two unit test programs should make it fairly straightforward to implement encoding, decoding, sending, and receiving of messages. Once that code works, implementing the actual application protocol is relatively easy, and fun!

Grading Criteria

Milestone 1:

- message encoding/decoding: 10%
- framing functions: 10%
- updater client implementation: 12.5%
- display client implementation: 12.5%
- design and coding style: 5%

Milestone 2:

- server implementation: 37.5%
- concurrency report: 7.5%
- design and coding style (server): 5%

Getting Started

To get started, download csf_assign05.zip and unzip it.

In the extracted `csf_assign05` directory, the `include` directory has header files and the `src` directory has C++ source files. The `build` directory is where compiled object files and executable files will be generated.

Generating header file dependencies (do this before compiling for the first time, or any time you add or change any `#include` directives):

```
make depend
```

To compile all executables:

```
make -j8
```

The `-j8` option tells `make` to use up to 8 processes to run commands, allowing you to take advantage of multiple CPU cores. You can adjust the number to adjust the degree of parallelism.

The `ref` directory contains the `server`, `updater`, and `display` executables compiled from the reference solution. They demonstrate the expected functionality, and you can use them to test your clients and server. (I.e., you can use the reference server to test your clients, and the reference clients to test your server.)

The following screencast video demonstrates running the programs, and how you can use the netcat program to emulate a client or server for testing purposes:

> https://jh.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=42dc6767-076e-47a4-8bf0-b42d01350c73

Overview

In this assignment, you will implement two network clients and a network server which together form a restaurant order system. The general idea is that this system could be used to keep track of current orders in a restaurant, encompassing point of sale (order entry), display (so the workers in the kitchen can see what needs to be prepared), and updating the status of orders (so that the back of house staff can see which items still need to be prepared, and deliver the food to the wait staff when the items are ready.)

The server maintains a collection of *orders*. Each order is a collection of *items*. Full details about orders and items are given in the Object Model section.

The *updater* client creates new orders and updates the status of existing items and orders. The *display* client shows the current state of all active orders and their constituent items.

The clients and server communicate with each other by sending *messages*. Full details about messages, their encoding, and the general network protocol are given in the Protocol and Encoding sections.

Restaurant Order System

This section documents the object model and network protocol to be implemented in the restaurant order system.

Object Model

The *object model* is the collection of data types representing orders, items, their statuses, and other important information. All of these types are declared in the `include/model.h` header file and defined in `src/model.cpp`.

`OrderStatus` is an enumeration type representing the status of an order. Its members are `OrderStatus::INVALID`, `OrderStatus::NEW`, `OrderStatus::IN_PROGRESS`, `OrderStatus::DONE`, and `OrderStatus::DELIVERED`. (Note that `OrderStatus::INVALID` is not a valid status, and is used only to represent the absence of a valid order status.)

`ItemStatus` is an enumeration type representing the status of an item within an order. Its members are `ItemStatus::INVALID`, `ItemStatus::NEW`, `ItemStatus::IN_PROGRESS`, and `ItemStatus::DONE`. (As with `OrderStatus`, the `ItemStatus::INVALID` member is not a valid status.)

The `Order` class represents an order and its constituent items. It has a unique integer identifier, and `OrderStatus` value, and a collection of item objects.

The `Item` class represents an item within an order. It has an order id (which is the unique id of the order the item is part of), an integer item id (which is unique within the overall order), an `ItemStatus` value, a description string, and an integer quantity (which must be positive).

The `Order` and `Item` classes have a variety of accessor functions for inspecting and modifying their data.

The `ClientMode` enumeration type has three members, `ClientMode::INVALID`, `ClientMode::UPDATER`, and `ClientMode::DISPLAY`. The last two are used to distinguish the two kinds of clients. When a client logs in, it includes the client mode it wants to use,

depending on which kind of client it is. (The `ClientMode::INVALID` value doesn't represent a valid client type, but it can be useful to indicate that the client mode is unknown.)

Messages

A *message* is a bundle of information sent from client to server (a "request") or from server to client (a "response"). The `Message` class, defined in `include/message.h`, represents one message.

The `MessageType` enumeration defines the various types of messages. These will be described in more detail in the Protocol section.

The `Message` class is designed to be able to represent any message. Each message type has a specific combination of data values it contains. So, the `Message` class's fields and accessor functions represent the union of all data values a single message could contain.

Protocol

The following table summarizes the message types, which program sends messages of that type, and what information a message of that type will contain.

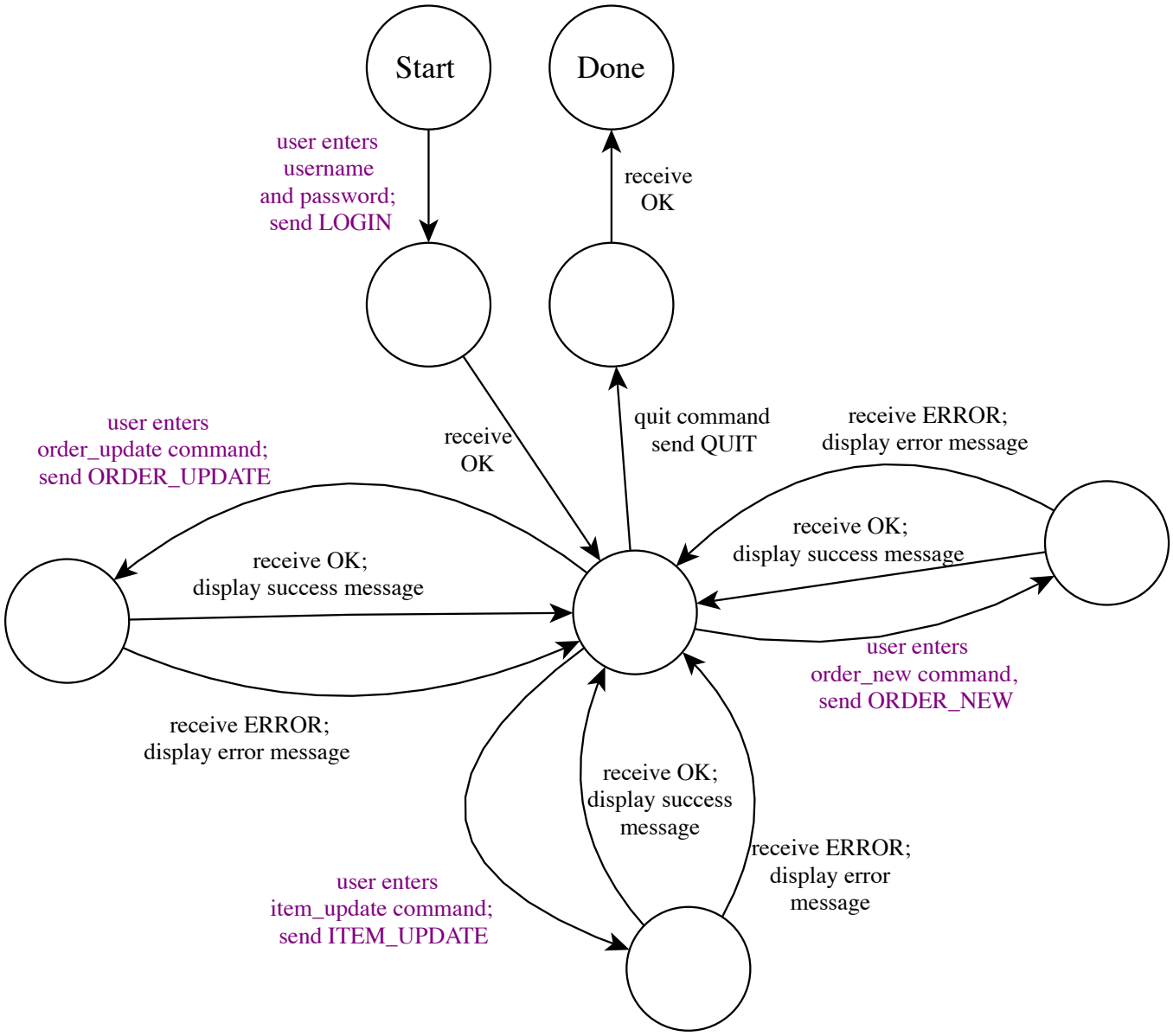
Message type	Sent by	Reeived by	Contained data values
`MessageType::LOGIN`	updater or display	server	client mode, string
`MessageType::QUIT`	updater	server	string
`MessageType::ORDER_NEW`	updater	server	order
`MessageType::ITEM_UPDATE`	updater	server	order id, item id, item status
`MessageType::ORDER_UPDATE`	updater	server	order id, order status
`MessageType::OK`	server	updater or display	string
`MessageType::ERROR`	server	updater or display	string
`MessageType::DISP_ORDER`	server	display	order
`MessageType::DISP_ITEM_UPDATE`	server	display	order id, item id, item status
`MessageType::DISP_ORDER_UPDATE`	server	display	order id, order status
`MessageType::DISP_HEARTBEAT`	server	display	*none*

The protocols implemented by the updater client, display client, and server are described by the following state machines. In each state machine, the nodes (circles) represent states, and the transitions (arrows) represent events. Each transition involves either sending or receiving a message. The "Start" node represents the initial state, and the "Done" node indicates that

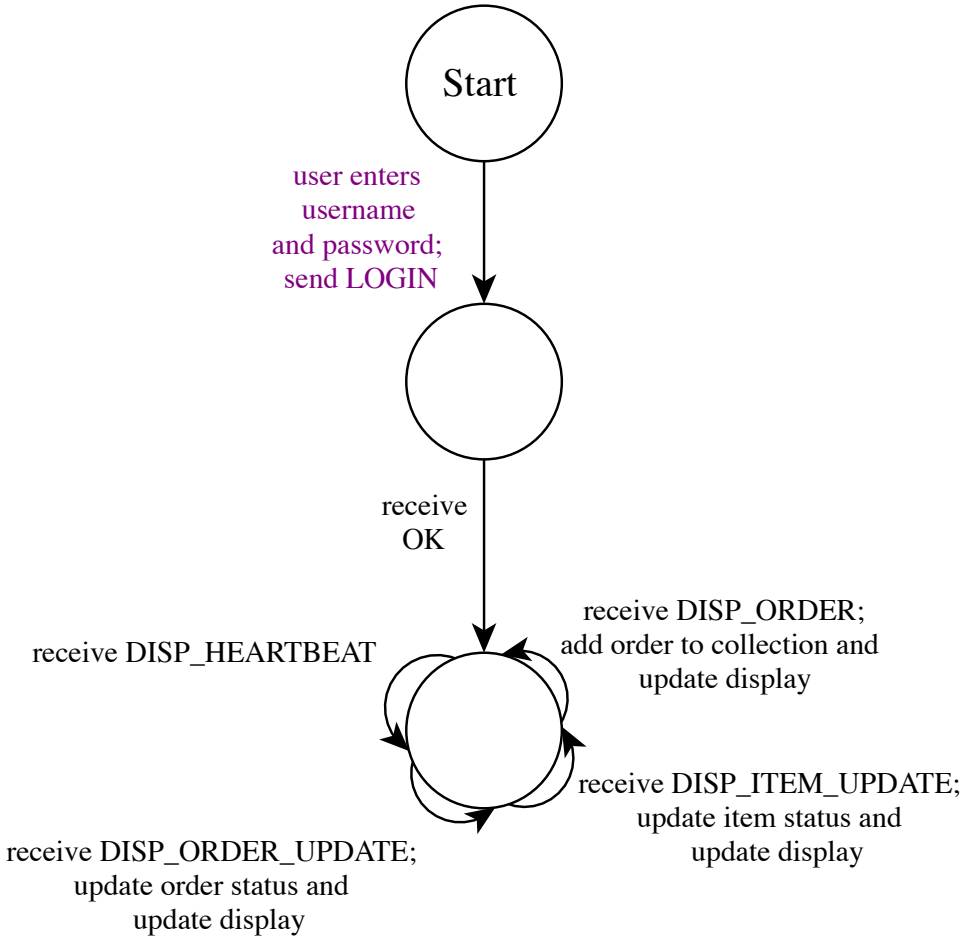
the conversation has finished and the network connection will be terminated.

Note that the edge labels in purple represent interactive commands that the user enters.

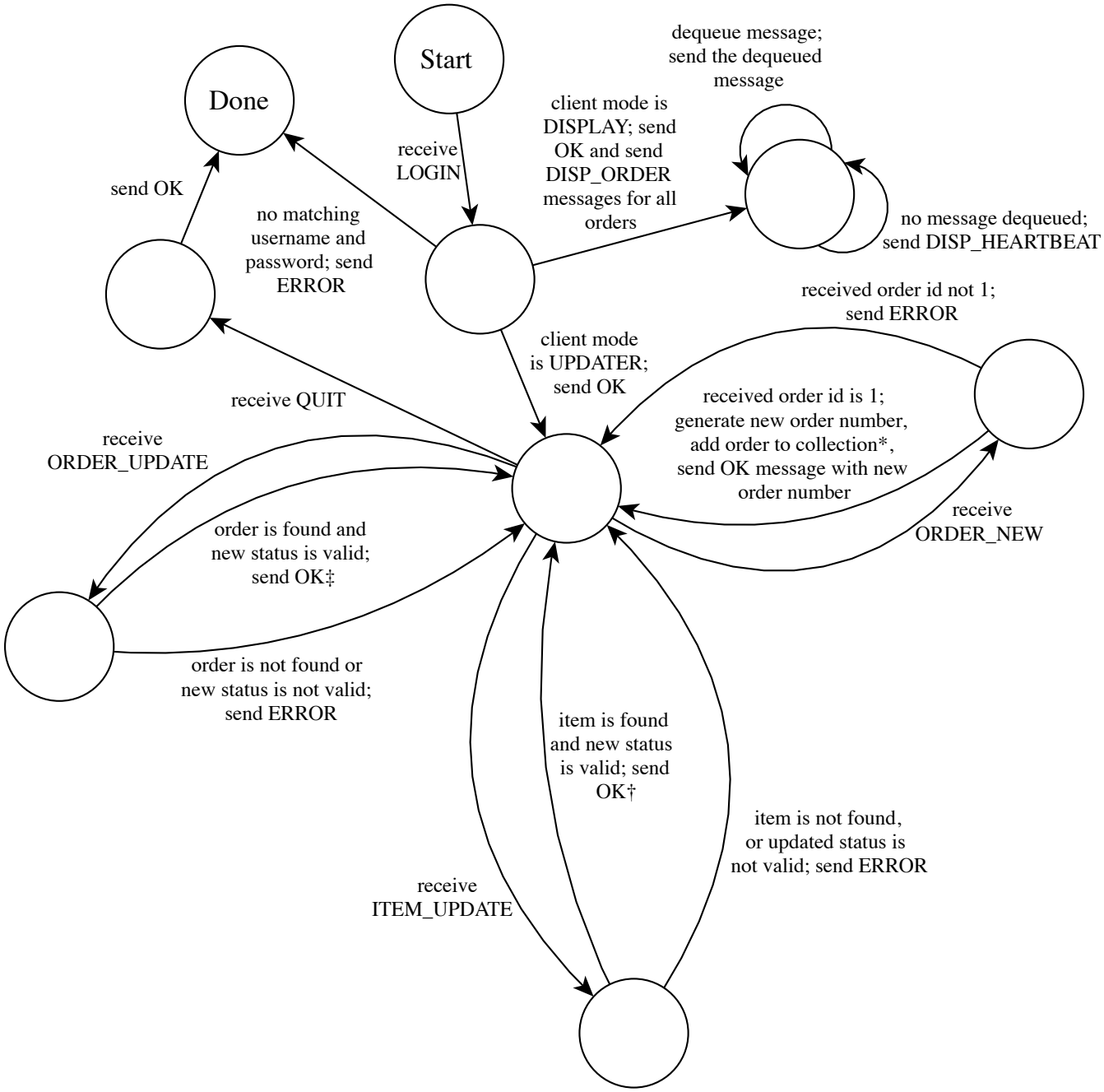
Updater state machine:



Display state machine:



Server state machine:



The server state machine describes the protocol implementing a conversation between the server and one client.

Note the following special cases in the server's state machines (indicated with the *, †, and ‡ symbols in the state diagram):

* When a new order is created and added to the collection, the server should enqueue a `MessageType::DISP_ORDER` message containing the order data to the message queues of each active display client.

† When a `MessageType::ITEM_UPDATE` message is successfully processed, the server should enqueue a `MessageType::DISP_ITEM_UPDATE` message containing the item id and new item status to the message queues of each active display client. Also, if as a result of applying the

item update, the order status transitions from `OrderStatus::NEW` to `OrderStatus::IN_PROGRESS`, or if the order status transitions from `OrderStatus::IN_PROGRESS` to `OrderStatus::DONE`, the server should enqueue a `MessageType::DISP_ORDER_UPDATE` message with the order id and new order status to the message queue of each active display client.

‡ When a `MessageType::ORDER_UPDATE` message is successfully processed, the server should enqueue a `MessageType::DISP_ORDER_UPDATE` message with the order id and new order status to the message queue of each active display client.

Encoding

In order to be sent and received via a TCP connection, a message is represented as a string, i.e., a sequence of bytes. The `Wire::encode` and `Wire::decode` functions (declared in `include/wire.h` and defined in in `src/wire.cpp`) implement conversion of a `Message` object to and from a string representation.

In the table of message types in the Protocol section, you will note that the last column is called "Contained data values". The entries in this column describe the "payload" of the message. The string representation of a message consists of the message type, followed by the contained data values (in order), with all items separated by a single "`|`" character.

A `MessageType` value can be converted to a string using the `Wire::message_type_to_str` function.

Integer data values such as order id and item id are encoded as a sequence of base 10 digits. You can use the `std::to_string` function to do this conversion.

`OrderStatus` and `ItemStatus` values can be converted to a string using (respectively) the `Wire::order_status_to_str` and `Wire::item_status_to_str` functions.

`MessageType::ORDER_NEW` and `MessageType::DISP_ORDER` messages contain an `Order` as the payload value. An `Order` is converted to a string consisting of the order id, order status, and item list, separacter by comma ("`,`") characters. The item list is a sequence of 1 or more items, separated by semicolon (";") characters. Each item is encoded as a string consisting of order id, item id, item status, description string, and integer quantity, each separated by colon ("`:`") characters.

Note that you may assume that the separator characters "`|`", "`,`", "`;`", and "`:`" will never occur in a string value within an encoded message.

The `message_tests` unit test program (`make build/message_tests`) has fairly comprehensive unit tests for message encoding and decoding. When you reach the point where all of the unit tests pass, you can have confidence that your implementations of `Wire::encode` and `Wire::decode` are working correctly.

Framing

Framing is the problem of determining where transmitted messages begin and end. The framing format for messages in the restaurant order system consists of a four-digit length

value, followed by an encoded message string, followed by a single newline ("`\n`") character. The length value specifies the number of bytes comprised by the encoded message string and the newline. For example, to frame the encoded message `OK|successful login`, the length value would be `0020`, since the encoded message is 19 characters long, and the newline contributes one extra character, for a total length of 20.

This framing scheme is an example of *run-length encoding*, which means that messages are preceded by the exact length of the message contents to follow, so that when receiving a message, the received can read the length (a known number of bytes), and then know exactly how many additional bytes to expect. Contrast this approach with a "terminating sentinel" style of framing, where the end of a message is indicated by a special sentinel character or character sequence.

The `IO::send` and `IO::receive` functions (declared in `include/io.h` and defined in `src/io.cpp`) frame and unframe a string value (i.e., an encoded message). `IO::send` writes the framed string to a file descriptor (e.g., a TCP socket), and `IO::receive` reads a framed string from a file descriptor. Note that these functions should throw `IOException` if any I/O error or EOF occurs.

The `io_tests` unit test program (`make io_tests`) tests the implementation of `IO::send` and `IO::receive`. Once these tests pass, you can be reasonably confident that your implementations of `IO::send` and `IO::receive` are working well.

Sending and Receiving Messages

Once you have the encoding and framing functions working, it is very easy to send and receive messages.

Let's say that a `Message` object `m` represents a message you want to send, and that `fd` is the file descriptor of the TCP socket connecting to the remote peer application. You can do so with the code

```
std::string s;
Wire::encode(m, s);
IO::send(s, fd);
```

Similarly, if you want to receive a message from the remote peer, the code would be something like

```
Message m;
std::string s;
IO::receive(fd, s);
Wire::decode(s, m);
// m now contains the received message
```

Implementation details

This section has further information about implementing the clients and server.

Shared Pointers

The model object classes (`Order`, `Item`, `Message`), and classes designed to be containers for model objects (e.g., `Message` and `MessageQueue`) consistently use `std::shared_ptr` to manage dynamically allocated objects. This approach has some important advantages:

- `std::shared_ptr` uses reference counting, and deletes the managed object when the last pointer to it goes out of scope; this is a simple form of garbage collection, and ensures that allocated objects won't be leaked
- because objects managed by `std::shared_ptr` are dynamically allocated, they are safe to transfer from one thread to another

If you want to create an object to be managed by `std::shared_ptr`, don't use the `new` operator, use the `std::make_shared` function. Its syntax is

```
std::make_shared<ClassName>(ConstructorArgs)
```

where *ClassName* is the name of the class you want the new managed object to be an instance of, and *ConstructorArgs* are any arguments you want to pass to the constructor of the new object. For example, if `order` is a `std::shared_ptr` managing an `Order` object, and you want to create a shared pointer to a new `Message` object that you can use to broadcast that order to display clients as a `MessageType::DISP_ORDER` message, you could use the code

```
auto order_new_msg =
  std::make_shared<Message>(MessageType::DISP_ORDER, order);
```

Note that shared pointers should be passed by value and returned by value if you need to pass them to or return them from functions.

The Updater Client

The updater client encapsulates both point-of-sale functionality (e.g., creating new orders) and back of house functionality (updating the status of items and orders.)

The updater client is invoked as

```
./build/updater HOSTNAME PORT
```

where `HOSTNAME` is the hostname or IP address the server is running on, and `PORT` is the TCP port number on which the server is listening for connections.

When the updater client starts, it should prompt the user to enter a username and password using the prompt text "`username: `" and "`password: `". (Note that there is a space after the colon in each prompt, and also that the program should not print a newline after the

prompt.)

The updater client should combine the entered username and password into a single credential string of the form "`username/password`", and send it to the server as a `MessageType::LOGIN` message. The server will respond with either a `MessageType::OK` or `MessageType::ERROR` message. If an ok response is received, the client continues to execute the command loop. Otherwise it prints an error message of the form

```
Error: error text
```

to `std::cerr` where *error text* is the string payload of the `MessageType::ERROR` message received from the server, and immediately exit with a non-zero exit code.

The command loop works as follows. The client prints the prompt "`> `" and reads a line of text, which is the command name. Commands are handled as follows:

`quit`: The client sends a `MessageType::QUIT` message to the server and receives the server's response. The server should send back a `MessageType::OK` message, and the client exits with an exit code of 0.

`order_new`: The client reads an integer number of items. Then, it reads exactly that many items. Each item is read by reading an integer item id, an item description string, and an integer quantity. The client then sends a `MessageType::ORDER_NEW` message containing an order with the specified items. The order id of the new order should be set to 1. The server will respond with a `MessageType::OK` message. The text in that message will have the form

```
Created order id OrderId
```

where *OrderId* is the actual order id assigned to the order.

`item_update`: The client reads an order id, item id, and item status. It sends a `MessageType::ITEM_UPDATE` message with the information entered. The server responds with a `MessageType::OK` or `MessageType::ERROR` message.

`order_update`: The client reads an order id and an order status. It sense a `MessageType::ORDER_UPDATE` message with the information entered. The server responds with a `MessageType::OK` or `MessageType::ERROR` message.

Note that when reading the additional values entered by the user for the `order_new`, `item_update`, and `order_update` commands, each value is read on a separate line, and the client should *not* print a prompt for any of the entered values. You should use `std::getline` to ensure that a complete line of text is read.

Note that there is no requirement to do anything special to handle invalid commands or input values. When testing your updater client, the autograder will only provide well-formed input.

For the `order_new`, `item_update`, and `order_update` commands, if the server responds

with `MessageType::OK`, the client should print a success message to `std::cout` of the form

```
Success: text
```

where *text* is the string payload of the received message. If the server responds with `MessageType::ERROR`, the client should print a failure message to `std::cout` of the form

```
Failure: text
```

where *text* is the string payload of the received message.

Note that for these commands (the commands other than `quit`) the command loop continues regardless of whether the response received was `MessageType::OK` or `MessageType::QUIT`.

If any I/O errors occur, if any invalid or incorrectly-formed message data is received, or if the server does not correctly implement the protocol as described in the Protocol section, the client should print an error message of the form

```
Error: explanation
```

to `std::cerr`, where *explanation* is any arbitrary text. In addition, if an error message is printed, the program should exit with a non-zero exit code to indicate failure.

Here is a transcript showing a run of the updater client, with user input in **bold**:

```
username: alice
password: foobar
> order_new
2
42
Veggie burger
1
101
Curly fries
2
Success: Created order id 1000
> item_update
1000
42
IN_PROGRESS
Success: successful item update
> item_update
1000
101
IN_PROGRESS
Success: successful item update
```

```
> item_update
1000
42
DONE
Success: successful item update
> order_new
1
67
Chocolate shake
1
Success: Created order id 1001
> quit
```

The Display Client

The display client implements a basic information display showing the status of all orders currently in the system.

It is invoked with the command

```
./build/display HOSTNAME PORT
```

where (as with the updater program) `HOSTNAME` is the hostname or IP address of the system where the server is running, and `PORT` is the TCP port the server is listening on.

The display client will prompt the user for a username and password in exactly the same way as the updater client. A login failure should be handled the same way as the updater client.

If the login is successful, the display client should clear the screen by printing the contents of the `CLEAR_SCREEN` string to `std::cout`. Note that you will need to flush the output buffer to make sure the contents of this string are sent right away, i.e.

```
std::cout << CLEAR_SCREEN << std::flush;
```

Once the server responds with the `MessageType::OK` response to the `MessageType::LOGIN` request, the display client should enter a loop where it receives messages from the server. These messages should be handled as follows:

`MessageType::DISP_ORDER`
: Add a new order to the collection of orders

`MessageType::DISP_ITEM_UPDATE`
: Update the status of a specific item within one of the current orders

`MessageType::DISP_ORDER_UPDATE`
: Update the status of a specific current order

`MessageType::DISP_HEARTBEAT`
: Do nothing; these messages are sent by the server only to detect display clients no longer connected

For all received messages other than `MessageType::DISP_HEARTBEAT` messages, after processing the received message, the client should clear the screen, and then refresh the display by printing the information in all orders. The orders should be printed in increasing order by order id. (Hint: using a `std::map` to manage the collection of current orders will make this easy.)

As a special case, if the display client receives a `MessageType::DISP_ORDER_UPDATE` message changing the status of an order to `OrderStatus::DELIVERED`, the order should be removed from the collection of orders before the display is refreshed. In other words, when the status of an order changes to delivered, it immediately disappears from the display.

The format for printing each order is as follows.

To begin printing an order, print (to `std::cout`) a line of the form

```
Order OrderId: OrderStatus
```

where *OrderId* is the order id, and *OrderStatus* is the order status.

Next, print each item, in the order in which the items appeared in the earlier `MessageType::DISP_ORDER` message which added the order to the display.

Printing an item consists of two lines. The first line has the form

```
  Item ItemId: ItemStatus
```

where *ItemId* is the item id, and *ItemStatus* is the item status, and the second line has the form

```
    ItemDescription, Quantity Qty
```

where *ItemDescription* is the item description string, and *Qty* is the item quantity.

Note that the first line of an item is preceded by two spaces, and the second line of an item is preceded by four spaces.

Note that once the username and password have been entered, the display client does not read any further user input. You can terminate a display client by typing Control-C in the terminal it's running in.

Here is what the display client would show in the terminal after the updates sent in the example session in the Updater Client section:

```
Order 1000: IN_PROGRESS
  Item 42: DONE
    Veggie burger, Quantity 1
  Item 101: IN_PROGRESS
    Curly fries, Quantity 2
Order 1001: NEW
```

```
  Item 67: NEW
    Chocolate shake, Quantity 1
```

The Server

For each client that connects to the server, the server should follow the server protocol according to the state diagram shown in the Protocol section.

You will need to implement the `Server::server_loop` member function so that it accepts TCP connections from clients, and for each one, starts a new thread to communicate with the client. Note that this member function does not return. The only way to terminate the server process is to send it a signal such as `SIGTERM`.

You should create a new instance of the `Client` class to manage the resources needed by a client thread. A pointer to the `Client` object should be passed to the new thread's start function as its argument.

The thread start function used to start new client threads should use `pthread_detach()` and `pthread_self()` to make the thread a detached thread, and then call the `Client::chat` member function of the thread's `Client` object. The `chat` member function should read requests from the client and send responses. Note that if the `chat` member function throws an exception, you should ensure that the thread is terminated gracefully, and that all resources (such as the TCP connection) are cleaned up. Assuming that the `Client` class's destructor does this cleanup, it should be sufficient to ensure that the `Client` object is deleted in order to ensure that resources are cleaned up.

When communicating with an updater client, the basic idea is to handle `MessageType::ORDER_NEW`, `MessageType::ITEM_UPDATE`, and `MessageType::ORDER_UPDATE` messages by updating the `Server` object's collection of `Order`s. Each time a new `Order` is created, a `MessageType::DISP_ORDER` message should be broadcast to all connected display clients. `MessageType::ITEM_UPDATE` and `MessageType::ORDER_UPDATE` messages should be handled by updating the appropriate `Item` or `Order`, and broadcasting a `MessageType::DISP_ITEM_UPDATE` or `MessageType::DISP_ORDER_UPDATE` message to active display clients.

Note the following special cases:

1. When the client sends a `MessageType::ORDER_NEW` message, the order id should be set to 1. The server should assign a new, valid order id. The valid order ids start at 1000, and increase by 1 with each new order.
2. When the first `Item` in an `Order` changes status from `ItemStatus::NEW` to `ItemStatus::IN_PROGRESS`, its `Order` changes status from `OrderStatus::NEW` to `OrderStatus::IN_PROGRESS`.
3. When all of the `Item`s in an `Order` have the status `ItemStatus::DONE`, their `Order` should change status to `OrderStatus::DONE`.
4. When a `MessageType::ORDER_UPDATE` message changes the status of an `Order` to `OrderStatus::DELIVERED`, the `Order` should be removed from the server's collection of orders.

Note that cases 2 and 3 mean that the server should broadcast both a `MessageType::DISP_ITEM_UPDATE` message and a `MessageType::DISP_ORDER_UPDATE` message. They should be broadcast in that order (`MessageType::DISP_ITEM_UPDATE` first, `MessageType::DISP_ORDER_UPDATE` second.)

Also note that for both items and orders, status updates must be applied in the correct sequence. Those sequences are as follows:

- For `ItemStatus`, the sequence is `NEW`, `IN_PROGRESS`, `DONE`
- For `OrderStatus`, the sequence is `NEW`, `IN_PROGRESS`, `DONE`, `DELIVERED`

When an updater client request can't be handled because the order or item it refers to doesn't exist, or when an order or item status change isn't valid, the server should send back a `MessageType::ERROR` response. We recommend using the `SemanticError` exception type to represent this situation.

When communicating with a display client, the `Client` object should wait for a shared pointer to a `Message` object to be enqueued on the `Client` object's `MessageQueue`. The idea is that when the server needs to broadcast display updates to all active display clients, it does so by enqueuing a `std::shared_ptr` to a `Message` to each display client's `MessageQueue`. If a shared pointer to a valid `Message` object can be dequeued within 1 second, that `Message` should be sent to the remote display client over the TCP connection. If no valid `Message` is available within one second, a `MessageType::DISP_HEARTBEAT` message should be send over the TCP connection. See the `MessageQueue`, Broadcasting to Display Clients for further details.

Another case to be aware of for display clients is that when a display client connects to the server, it should be sent `MessageType::DISP_ORDER` messages for all current orders.

For both updater and display clients, if there an `IOException` occurs when receiving or sending a message, the client thread should immediately cease communicating with the client, clean up resources, and terminate the thread.

Synchronization of Shared Data

The single `Server` object should maintain all of the data about `Orders`, `Items`, and their statuses. Each `Client` object as a pointer to the `Server` object. You should add fields and member functions to the `Server` class that the `Client` object can use to have the `Server` do operations on the order and item data. Also, the `Server` will need to be able to broadcast messages to active display clients, so you will need to implement a way for the `Server` instance to keep track of active display clients.

You will need to synchonize access to any data visible to multiple threads. In general, a `pthread_mutex_t` is sufficient to control access to shared data in situations where the only concern is preventing race conditions, and none of the operations any thread will be performing will involve waiting for a condition to be true.

`MessageQueue`, Broadcasting to Display Clients

As we've discussed in class, queues are a great way to implement a communication channel

from one thread to another. The `MessageQueue` class is intended to allow the server to send (shared) pointers to `Message` objects to active display clients. `MessageQueue` is not intended to be a bounded queue, so there is no enforced limit on how many messages the server can add to a display client's `MessageQueue`.

You will need to implement a mechanism that allows the client thread invoking the `dequeue()` operation to remove message from the queue to wait until either

1. The queue contains at least one `Message`, or
2. The queue remains empty for one second

We suggest using a `sem_t` (semaphore) object to keep track of how many messages have been added to the queue. Initially, the semaphmore count should be 0. The `enqueue()` operation should use `sem_post` to increment the semaphore, indicating that there is now one more message in the queue. The `dequeue()` member function can use `sem_timedwait` to wait for the queue to be nonempty. If you pass a `timespec_t` value to `sem_timedwait` set for one second in the future, `sem_timedwait` will return with an error if the semaphore count remains 0 for one second. You can use the following code to initialize a `timespec_t` value for a time one second in the future:

```
std::timespec ts;
std::timespec_get(&ts, TIME_UTC);
ts.tv_sec += 1; // wait for one second
```

You can read more about `sem_timedwait` using the command `man sem_timedwait`.

When the server needs to broadcast a message to all active display clients, it should iterate over the collection of active display clients, and use `MessageQueue::enqueue` to add a copy of the message to each display client's message queue. Note that you can use `Message::duplicate` to return a shared pointer to an exact copy of a specified `Message`. Giving each display client thread a shared pointer to a distinct dynamically allocated `Message` object is a good idea; allowing threads to access the same object instance can lead to conflicts, so avoiding unnecessary sharing of objects is a good practice for multithreaded programming.

Synchronization Report

In your `README.txt` for Milestone 2, you should write a brief report explaining how you used synchronization to ensure that your server has no race conditions or deadlocks. Your report should explicitly indicate what data is shared between threads, and how that data is synchronized.

Exceptions, RAII

We highly recommend using exceptions to deal with any exceptional circumstance that means that control cannot continue normally in the program.

Some useful exception types are defined in `include/except.h`.

`Wire::decode`, `IO::send`, and `IO::receive` are required to throw `InvalidMessage` and `IOException` exceptions as appropriate. When any program (client or server) encounters an improperly-formed or improperly-framed message, or when an I/O error or EOF condition occurs when reading from or writing to a TCP socket, the program should immediately cease communication with the remote peer. By letting these exceptions propagate naturally, you should be handle them with a single high-level `try`/`catch` construct.

The `ProtocolError` exception type is meant to represent a situation where a properly-formed message was received, but it violates the protocol as defined by the relevant state machine. When a remote peer violates the protocol, the program should immediately cease communication with the remote peer.

The `SemanticError` exception type is intended to be used by the server for situations where a message was properly formed, and did not violate the protocol state machine, but the operation embodied by the message was not semantically correct. For example, an order cannot have its status changed unless its status is `OrderStatus::DONE`, and the only valid order update that can be requested by an updater client is to change the order status from `OrderStatus::DONE` to `OrderStatus::DELIVERED`. If a member function in the `Server` class detects that these rules have been violated, it can throw `SemanticError`. This is a useful exception type because it can be caught in order to detect that the operation requested by the client is invalid, and the server can send back a `MessageType::ERROR` message in response.

"RAII" stands for "Resource Acquisition Is Initialization", and is a C++ philosophy that advocates using a scoped local object to ensure the release of a resource when it is no longer needed. Because destructors for local objects are called regardless of how control leaves a scope, they can ensure that a resource is cleaned up even an exception is thrown. In general, if your program uses exceptions, it should also use RAII consistently to clean up resources.

`std::unique_ptr` is useful for implementing RAII for a dynamically allocated object. For example, in the server, you should pass a pointer to a dynamically allocated `Client` object to the thread start function of the thread tasked with communicating with a client, in order to give the thread the access to the resources it needs. A `std::unique_ptr` is a useful way to ensure that this object gets cleaned up before the thread exist. A mutex guard object implements RAII for a critical section, ensuring that the mutex is released. For the client programs, you might find it useful to use RAII to ensure that the client file descriptor gets closed before the program terminates.

Submitting

You can create a zipfile of your work by running the command

```
make solution.zip
```

To submit, upload your `solution.zip` file to **Assignment 5 MS1** or **Assignment 5 MS2** on Gradescope, depending on which milestone you are submitting.

Keep in mind that for Milestone 2, your `README.txt` should contain your synchronization report.